

Boss Challenge — Paper 17 (Class 7)

Respiration in Organisms | Lines & Angles | Electric Current & its Effects | Exponents & Powers
One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. The main job of respiration inside a living cell is to break down food in order to:
(A) release energy
(B) remove all water
(C) cool the body down
(D) make more food
2. Two angles whose measures add up to exactly 90° are called:
(A) vertically opposite angles
(B) adjacent angles
(C) supplementary angles
(D) complementary angles
3. An electric current can flow in a circuit only when the path is:
(A) a single straight wire with no cell
(B) closed (complete)
(C) made only of plastic
(D) open at the switch
4. In the power 2^5 , the small raised number 5 tells us how many times the base 2 is:
(A) subtracted from itself
(B) multiplied by itself
(C) divided by itself
(D) added to itself
5. During aerobic respiration the body uses up a gas taken in from the air. That gas is:
(A) oxygen
(B) hydrogen
(C) carbon dioxide
(D) nitrogen
6. Two angles whose measures add up to exactly 180° are called:
(A) corresponding angles
(B) supplementary angles
(C) acute angles
(D) complementary angles
7. The device used to open or close a circuit, switching a current on or off, is the:
(A) switch
(B) wire
(C) bulb
(D) cell

8. Evaluate the cube of two. Worked out fully, the value of 2^3 comes to:
- (A) 9
 - (B) 8
 - (C) 23
 - (D) 6
9. The gas that a person breathes OUT in a larger amount than they breathe in is:
- (A) oxygen
 - (B) water vapour only
 - (C) nitrogen
 - (D) carbon dioxide
10. The complement of an angle of 35° (the angle you add to make 90°) is:
- (A) 35°
 - (B) 55°
 - (C) 145°
 - (D) 65°
11. When an electric current passes through a thin wire, the wire becomes hot. This is called the:
- (A) lighting effect of current
 - (B) heating effect of current
 - (C) magnetic effect of current
 - (D) chemical effect of current
12. Written as a power of 10, the number 10 000 (ten thousand) is:
- (A) 10^4
 - (B) 10^5
 - (C) 10^3
 - (D) 4^1
13. A simple test in the lab to show that exhaled air contains carbon dioxide is that it turns:
- (A) lime water milky
 - (B) blue litmus red
 - (C) starch blue-black
 - (D) a glowing splint bright
14. The supplement of an angle of 110° (the angle you add to make 180°) is:
- (A) 250°
 - (B) 70°
 - (C) 80°
 - (D) 110°

15. The coil inside an electric heater or toaster is made of a special wire that gives out a lot of heat.

This wire is:

- (A) nichrome
- (B) aluminium foil
- (C) plastic
- (D) copper

16. Using the rule $a^m \times a^n = a^{m+n}$, the product $2^2 \times 2^3$ equals:

- (A) 4^2
- (B) 2^5
- (C) 2^1
- (D) 2^6

17. The tiny structures inside a cell where most energy is released are nicknamed the powerhouse.

They are the:

- (A) mitochondria
- (B) chloroplasts
- (C) cell walls
- (D) vacuoles

18. When two straight lines cross, the pair of angles directly opposite each other at the crossing are always:

- (A) always 90°
- (B) supplementary
- (C) complementary
- (D) equal

19. A safety device that contains a thin wire which MELTS and breaks the circuit when the current grows too large is a:

- (A) bulb
- (B) fuse
- (C) cell
- (D) switch

20. Using the rule $a^m / a^n = a^{m-n}$, the quotient $5^3 / 5^2$ equals:

- (A) 5^1
- (B) 5^3
- (C) 1^3
- (D) 5^5

21. When there is not enough oxygen, yeast carries out anaerobic respiration and produces alcohol along with the gas:

- (A) nitrogen
- (B) carbon dioxide
- (C) hydrogen
- (D) oxygen

22. Two adjacent angles formed on a straight line (a linear pair) always add up to:

- (A) 270°
- (B) 90°
- (C) 180°
- (D) 360°

23. A compass needle placed near a wire moves when current flows through the wire. This shows that an electric current has a:

- (A) heating effect
- (B) no effect at all
- (C) magnetic effect
- (D) chemical effect

24. Any non-zero number raised to the power 0 is equal to:

- (A) undefined for all numbers
- (B) the base itself
- (C) 1
- (D) 0

25. After running very fast, the muscles in your legs may respire without enough oxygen and build up:

- (A) alcohol
- (B) lactic acid
- (C) glucose
- (D) starch

26. An angle that measures more than 90° but less than 180° is called:

- (A) a reflex angle
- (B) a right angle
- (C) an acute angle
- (D) an obtuse angle

27. A coil of wire wound around an iron core becomes a magnet ONLY while current flows. This device is an:

- (A) simple cell
- (B) permanent magnet
- (C) electromagnet
- (D) fuse

28. The number 64 can be written neatly as a power of 2. It is equal to:

- (A) 6^2
- (B) 2^6
- (C) 2^8
- (D) 2^{16}

29. Humans take air into a pair of spongy organs in the chest. These breathing organs are the:
- (A) kidneys
 - (B) liver
 - (C) gills
 - (D) lungs
30. An angle that measures more than 180° but less than 360° is called:
- (A) a reflex angle
 - (B) a complete angle
 - (C) an obtuse angle
 - (D) a straight angle
31. An electric bell rings repeatedly. It works using the:
- (A) chemical effect of current
 - (B) lateral inversion
 - (C) magnetic effect of current
 - (D) heating effect of current
32. Using the rule $(a^m)^n = a^{mn}$, the value of $(3^2)^3$ is:
- (A) 3
 - (B) 3
 - (C) 3
 - (D) 9^3
33. A fish stays underwater all its life and takes in oxygen dissolved in the water using its:
- (A) gills
 - (B) lungs
 - (C) spiracles
 - (D) skin only
34. The line that crosses two or more other lines at distinct points is called a:
- (A) perpendicular only
 - (B) parallel line
 - (C) ray
 - (D) transversal
35. The thin coiled wire inside an electric bulb that glows brightly when current passes is called the:
- (A) terminal
 - (B) fuse
 - (C) electrode
 - (D) filament
36. Comparing the two numbers 3^4 and 4^3 , which one is LARGER?
- (A) they are equal
 - (B) 4^3 , because 4 is bigger than 3
 - (C) 4^3 , because it equals 81
 - (D) 3^4 , because it equals 81 while 4^3 equals 64

37. An earthworm has no lungs or gills. It exchanges gases for respiration through its:
- (A) dry scales
 - (B) feathery gills
 - (C) a pair of lungs
 - (D) moist skin
38. When a transversal cuts two parallel lines, each pair of corresponding angles is:
- (A) equal
 - (B) supplementary
 - (C) complementary
 - (D) always right angles
39. Two or more cells joined together end to end so they can drive a current make a:
- (A) fuse
 - (B) switch
 - (C) battery
 - (D) filament
40. The value of (-1) raised to any EVEN power, for example $(-1)^4$, is:
- (A) 0
 - (B) 4
 - (C) +1
 - (D) -1
41. Insects such as a cockroach take air directly into their bodies through tiny breathing holes called:
- (A) gills
 - (B) stomata
 - (C) spiracles
 - (D) alveoli
42. When a transversal cuts two parallel lines, a pair of co-interior (allied) angles on the same side adds up to:
- (A) 180°
 - (B) 360°
 - (C) they are equal
 - (D) 90°
43. If more cells are added in series in a circuit (without overloading it), the bulb will usually glow:
- (A) brighter
 - (B) with no change
 - (C) dimmer
 - (D) a different colour

44. Taking care of the sign, the cube of negative two, written $(-2)^3$, has the value:

- (A) -6
- (B) +8
- (C) +6
- (D) -8

45. A green plant does not have a nose. The tiny pores on its leaves through which gases pass are the:

- (A) roots only
- (B) stomata
- (C) xylem tubes
- (D) the bark

46. Two angles form a linear pair. If one of them is 70° , the other angle must be:

- (A) 20°
- (B) 70°
- (C) 110°
- (D) 290°

47. An electromagnet can be made STRONGER by increasing the current and also by:

- (A) using a wooden core
- (B) removing the cell
- (C) painting the wire
- (D) winding more turns of wire in the coil

48. Written in standard (scientific) form, the number 3 450 000 is:

- (A) 3.45×10^6
- (B) 34.5×10^5
- (C) 3.45×10^5
- (D) 3.45×10^4

49. When you breathe IN, a sheet of muscle below your lungs flattens and moves down. This muscle is the:

- (A) the ribs alone
- (B) the heart
- (C) the windpipe
- (D) diaphragm

50. Two lines crossing make four angles. If one angle is 65° , the angle vertically opposite to it is:

- (A) 25°
- (B) 90°
- (C) 115°
- (D) 65°

51. An electric iron, a room heater and a toaster all work mainly because of the:

- (A) heating effect of electric current
- (B) reflection of light
- (C) chemical effect of electric current
- (D) magnetic effect of electric current

52. The value of 5^2 (five squared) is:

- (A) 25
- (B) 10
- (C) 7
- (D) 52

53. Taking air into the body and giving it out is called breathing, while the release of energy from food is called:

- (A) digestion
- (B) transpiration
- (C) respiration
- (D) circulation

54. The special angle that is exactly equal to its own complement is:

- (A) 60°
- (B) 90°
- (C) 45°
- (D) 30°

55. The symbol for an electric cell shows two lines: a longer thin line and a shorter thick line. The longer thin line stands for the:

- (A) negative terminal
- (B) positive terminal
- (C) switch
- (D) fuse

56. Using the rule $(a \times b)^n = a^n \times b^n$, the value of $(2 \times 3)^2$ is:

- (A) 12
- (B) 25
- (C) 18
- (D) 36

57. Bubbles of gas trapped in rising bread dough come from yeast. As the dough is baked, this makes the bread:

- (A) hard and solid
- (B) flat with no rise
- (C) soft and full of holes
- (D) sour and wet

58. Two lines in a plane that always stay the same distance apart and never meet, however far they go, are called:

- (A) parallel lines
- (B) concurrent lines
- (C) intersecting lines
- (D) perpendicular lines

59. A fuse wire is chosen so that it melts easily at the right moment. Compared with ordinary wire, a fuse wire has a:

- (A) very high melting point
- (B) low melting point
- (C) no resistance at all
- (D) a magnetic core

60. The sum $7\blacksquare + 3\blacksquare$ is equal to:

- (A) 1
- (B) 2
- (C) 0
- (D) 10

61. The roots of a land plant also need to breathe. They get the oxygen they use from the:

- (A) sunlight on the leaves
- (B) water rising in the stem
- (C) air spaces between soil particles
- (D) carbon dioxide in the soil

62. The sum of all the angles formed around a single point (going once all the way round) is:

- (A) 360°
- (B) 180°
- (C) 270°
- (D) 90°

63. If the filament of a bulb breaks, the bulb stops glowing because the:

- (A) cell has become a magnet
- (B) bulb has run out of light
- (C) glass has changed colour
- (D) circuit is no longer complete

64. Written as a power of 10, the number 1000 (one thousand) is:

- (A) 10^3
- (B) $3^1\blacksquare$
- (C) $10\blacksquare$
- (D) 10^2

65. Aerobic respiration releases much more energy than anaerobic respiration because aerobic respiration:

- (A) works only in plants
- (B) makes alcohol as well
- (C) uses oxygen to break glucose down completely
- (D) needs no food at all

66. Two complementary angles are exactly equal to each other. Each of these angles measures:

- (A) 30°
- (B) 60°
- (C) 45°
- (D) 90°

67. Hans Christian Oersted noticed a compass needle move near a wire carrying current. His discovery linked electricity with:

- (A) light
- (B) sound
- (C) heat only
- (D) magnetism

68. The value of 2^4 (two to the power four) is:

- (A) 32
- (B) 6
- (C) 16
- (D) 8

69. Gas exchange in the human lungs happens at millions of tiny, balloon-like air sacs called:

- (A) spiracles
- (B) villi
- (C) nostrils
- (D) alveoli

70. An angle that measures exactly 90° is known as a:

- (A) straight angle
- (B) right angle
- (C) reflex angle
- (D) acute angle

71. In a circuit diagram, the gap in the line with a small lever that can be joined or left open represents a:

- (A) switch
- (B) cell
- (C) bulb
- (D) resistor / coil

72. The value of 3^3 (three cubed) is:

- (A) 33
- (B) 12
- (C) 27
- (D) 9

73. A frog is special because, besides its lungs, it can also take in oxygen through its:

- (A) gills throughout life
- (B) dry feathers
- (C) a shell
- (D) moist skin

74. When a transversal crosses two parallel lines, a pair of alternate interior angles (the Z-shape) are:

- (A) supplementary
- (B) always 90°
- (C) equal
- (D) complementary

75. An MCB (Miniature Circuit Breaker) is used in modern homes instead of a fuse because, unlike a fuse, an MCB:

- (A) never breaks the circuit
- (B) makes the current larger
- (C) gives out light when working
- (D) can be switched back on after it trips

76. Any number raised to the power 1 (for example 9^1) is simply equal to:

- (A) the number itself
- (B) 1
- (C) double the number
- (D) 0

77. FUSION: A single yeast cell respires and DOUBLES into two every hour. Starting from one cell, the number after 5 hours is best written as the power:

- (A) 2 to the power 4, which equals 16
- (B) 2 to the power 5, which equals 32
- (C) 5 to the power 2, which equals 25
- (D) 2 times 5, which equals 10

78. The complement of any acute angle (an angle smaller than 90°) is always:

- (A) another acute angle
- (B) a reflex angle
- (C) a right angle
- (D) an obtuse angle

79. Why is copper, and not nichrome, used for the long connecting wires that join the parts of a circuit?

- (A) copper lets current pass easily without heating up much
- (B) copper is magnetic
- (C) copper melts to protect the circuit
- (D) copper has a high melting point so it glows

80. The number 10^6 (ten to the power six) written out in full is:

- (A) one thousand
- (B) sixty
- (C) one million (1 000 000)
- (D) one lakh

81. FUSION: At rest a child breathes 18 times each minute. The total number of breaths taken in a quarter of an hour (15 minutes) is:

- (A) 180 breaths
- (B) 33 breaths
- (C) 270 breaths
- (D) 1080 breaths

82. Two angles are supplementary and one of them is a right angle (90°). The other angle is therefore:

- (A) 180°
- (B) 45°
- (C) also 90°
- (D) 0°

83. FUSION: A toy's holder has slots for cells, and the slots are arranged in 2 rows with 3 cells in each row, so the total is $2 \times 2 \times \dots$ no - the toy actually has 2^3 cells in all. The number 2^3 equals:

- (A) 23 cells
- (B) 5 cells
- (C) 6 cells
- (D) 8 cells

84. FUSION: A bacterium SPLITS into 2 every 20 minutes. Starting from a single bacterium, after 4 splits the count is best written as the power:

- (A) 4^2 , which equals 16
- (B) 2^4 , which equals 16
- (C) 2×4 , which equals 8
- (D) 2^3 , which equals 8

85. We start to breathe faster and more deeply during exercise mainly because the working muscles:

- (A) need extra oxygen to release more energy
- (B) have stopped respiring
- (C) are trying to cool down
- (D) need to take in carbon dioxide

86. FUSION: A current-carrying wire deflects a compass needle so that it turns to 50° from its rest line. The angle still needed to complete a right angle (90°) from the rest line is:

- (A) 50°
- (B) 40°
- (C) 130°
- (D) 90°

87. FUSION: An electromagnet's strength is tested by counting paper clips it can lift. The strongest electromagnet here lifts 10^2 clips. The number 10^2 equals:

- (A) 1000 clips
- (B) 20 clips
- (C) 100 clips
- (D) 12 clips

88. FUSION: A heater coil's resistance, in tiny units, is given as 10^3 . Written as an ordinary number, 10^3 is:

- (A) 30
- (B) 1000
- (C) 100
- (D) 10000

89. Cells that need a great deal of energy, such as busy muscle cells, contain a large number of:

- (A) cell walls
- (B) vacuoles
- (C) nuclei
- (D) mitochondria

90. FUSION: A switch lever swings through a straight angle (a half-turn) as it is flicked. The number of degrees it sweeps through is:

- (A) 180°
- (B) 45°
- (C) 90°
- (D) 360°

91. A device that uses the heating effect to make a wire glow and so give out light is best described as a:

- (A) bulb
- (B) cell
- (C) fuse
- (D) compass

92. Using the rule for the same base, $4^1 \div 4^1$ equals:

- (A) 4^1
- (B) 1
- (C) 0
- (D) 4

93. When you breathe OUT, the diaphragm relaxes and the chest cavity becomes smaller. This pushes the air out because the:

- (A) diaphragm flattens down
- (B) space for the lungs shrinks
- (C) ribs swing further out
- (D) lungs fill with more air

94. Two adjacent angles must, by definition, share a common vertex and one common:

- (A) length
- (B) midpoint
- (C) circle
- (D) arm (side)

95. To make a simple electromagnet you should wind the coil of wire around a core made of:

- (A) wood
- (B) soft iron
- (C) rubber
- (D) glass

96. The value of 2^{10} (two to the power ten), a number that often appears with computers, is:

- (A) 512
- (B) 1024
- (C) 20
- (D) 100

97. Tiny living things (microbes) in the soil and the air also respire. This tells us that respiration is a process carried out by:

- (A) all living organisms
- (B) only green plants
- (C) only large animals
- (D) only animals that move

98. If two angles are both equal AND supplementary, then each of them must measure:

- (A) 60°
- (B) 45°
- (C) 180°
- (D) 90°

99. The two metal ends of an electric cell, to which the connecting wires of a circuit are joined, are called its:

- (A) fuses
- (B) filaments
- (C) terminals
- (D) switches

100. The number 10^5 (ten to the power five) written out in full, the same as one lakh, is:

- (A) 10,000
- (B) 10,00,000
- (C) 50
- (D) 1,00,000